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Study of automatic prediction of emotion from handwriting samples

Michael Fairhurst, Meryem Erbilek, Cheng Li

School of Engineering and Digital Arts, University of Kent, Canterbury, Kent CT2 7NT, UK
E-mail: M.C.Fairhurst@kent.ac.uk

Abstract: Handwriting biometrics has a long history, especially when the handwritten signature is the target, but it has also proved possible to use handwriting as a basis for the prediction of various non-unique but forensically useful characteristics of the writer, generally considered to be examples of so-called ‘soft biometrics’. Most commonly, these are characteristics such as the age or gender of the writer, but the predictive capabilities arising in handwriting offer wider opportunities for trait prediction. This study presents a preliminary investigation of the use of handwriting to predict information about the writer relating specifically to higher level characteristics such as emotional state. The authors present an initial study to demonstrate that this is possible, and explore a number of factors particularly relevant to the use of such a capability in areas of forensic investigation.

1 Introduction

It is becoming increasingly apparent that the fields of biometrics and forensics can benefit considerably from establishing closer links which cross their traditional boundaries. An obvious example is the way in which the study of soft biometrics has been developing. Although traditional biometric systems utilise information which is assumed to be unique to an individual, thus allowing the possibility of identifying the ‘owner’ of the data, soft biometrics are explicitly known not to be unique, but are nevertheless characteristic of an individual, and therefore can also be a contributory source of supplementary identification information [1]. Among the most commonly encountered soft biometrics are general characteristics such as subject age or gender, which provide important information about an individual – perhaps especially in a typical forensics scenario – without providing a specific identification label. Sometimes more specific traits are helpful, such as in handwriting biometrics where the determination of ‘handedness’ might be important. Although falling short of enabling an absolute identity decision, such measures at the very least allow elimination of some possible individuals from consideration in an identification scenario, narrowing the required search space, and could thus be of considerable potential importance in forensic investigations.

Many examples can be found of studies which utilise soft biometrics to improve the accuracy of the identification process or to refine a list of likely identities, but there is also an expanding literature reporting studies which attempt to predict soft biometric characteristics from conventional biometric measurements [2, 3]. To date, however, such predictive studies have been confined to a small number of characteristics, such as those mentioned above. Yet an extension of predictive capabilities to higher levels of abstraction might provide even more powerful tools to support a range of practical scenarios, including forensics, but also beyond this, for example in healthcare scenarios and elsewhere. These higher level predictions might typically relate to conditions broadly describable as mental and emotional state in varying forms (here we will use the latter term, for consistency in comparisons with other published work, although we recognise that other studies may adopt stricter and more formal definitions). For example, a knowledge of whether at a particular time an individual is happy or sad, anxious or calm, under stress or relaxed and so on, might all provide information which could be extremely valuable in interpreting particular scenarios or evaluating human activities. To illustrate this, if a fragment of handwriting can reveal, on analysis, a high likelihood that the writer was under stress at the time of writing, this could be valuable information for those investigating a related crime. Although our proposed techniques could not distinguish between different sources of stress (e.g. general self-imposed pressure and explicit threat or coercion), they can provide an objective and reproducible metric of degree of stress being experienced, and can thus generate important, even though not necessarily absolute, evidential support. We should note also in passing that techniques in forensics can similarly stimulate new approaches in biometrics – for example, in identifying more powerful or appropriate features to characterise human activity – and so the disciplines of biometrics and forensics can be mutually supportive.

In this paper we report an experimental study to explore the possibility of developing a predictive capacity at this level, based on data in the form of handwriting fragments.

Although this is a preliminary study, we believe it to be significant both because we are not aware of any directly comparable study in the literature, and in the sense that it identifies some of the important relevant issues which need to be addressed further if such work is to be developed robustly in the future. It also provides some indications of what might be achievable with further work. Our starting point is to observe that the most likely biometric modalities to offer the potential for this 'higher state' prediction are the behavioural modalities rather than the physiological, for obvious reasons, and in this study we restrict ourselves to just one illustrative modality (specifically handwriting). This paper is a revised, updated and more fully developed version of [4], invited for submission following the IWBF 2014 Workshop, and includes many new results not previously reported in the original paper.

2 Background

Reported analyses of predicting human emotions from biometrics have mostly been investigated with respect to the face modality [5–7], but also for the voice [8], gait [9] and keystroke [10] modalities.

Although, particularly within the handwriting biometric research community, substantive studies of the analysis of emotion prediction are difficult to find, there are some related studies which suggest that prediction of emotion from handwriting biometrics may be possible. For instance, the study reported in [11] focused on determining a particular personality characteristic – specifically, emotional control – of an individual from his/her handwriting. This particular study reported a way to recognise the level of the emotion control (very low, low, medium, high or very high) of a person by using features related to the baseline or the slope line of handwriting, and applying a set of fuzzy rules. 30 handwriting samples (20 samples for training and 10 samples for the testing phase) were used for the experimental study. Beyond this type of study, it is true that work by graphologists can be found which points to links between handwriting and personality traits, but little of this work is published in the technical scientific literature and, in any case, typically relies primarily on human analysis. Some more objectively documented work can be found in the literature (see e.g. [12]) but this is not directly relevant to the study we report here.

More extensive but more narrowly targeted work can be found which demonstrates the biomedical potential of adopting a biometric analysis of signatures for developing systems which can discriminate specific health conditions (e.g. hypertension, hypercholesterolemia, and cardiac problems). This approach is explored particularly in [13], in which 120 subjects performed nine neuromuscular tests, and during each acquisition between 5 and 7 samples of signatures were collected. The results obtained showed that several such conditions seem to be predictable to some extent.

A considerable amount of other reported work has been very successful in using handwritten expression (often based on hand-executed drawings rather than writing as such) as an aid to detecting or assessing neuropsychological and similar conditions, and good examples can be found in relation to visuo-spatial neglect post-stroke, developmental coordination disorder in children, Parkinson's disease and so on [14–16].

From a rather different perspective, in [17], the authors propose a fundamental theoretical framework of eight principles to detect lies based on information arising from handwriting expression, but it is difficult to predict how successful this methodology might be for automated pattern recognition algorithms.

However, there is little formally reported work which directly achieves successful prediction of emotion/mental state, in the sense described in Section 1, based on the automated analysis of handwriting. In this paper, therefore, we report a preliminary study which enables us to move away from almost exclusively human inspection of handwriting as a means of identifying higher level emotional states, and instead provides pointers to how we can develop an automated capability for predicting from a handwriting sample the emotional state of the writer at the time of sample execution. Our aim, in what we thus believe is the first study of its kind, is to demonstrate that objective metrics of emotional state can be extracted from handwriting through an automated process, providing objective indicators of human characteristics at a given time and under varying acquisition conditions. We also aim to provide some clear indications about how such work might usefully be developed and refined in the future.

3 Methodology

A fundamental requirement for evaluating any proposed technique is the availability of data tagged with relevant ground truth information. In the area addressed by our study, there is no publicly available database, and hence a principal contribution of our work was to generate an appropriate database, an extensive and very time-consuming exercise in itself. We achieved this by means of a small set of exercises which allowed us to acquire information from participants in both 'fixed' tasks (i.e. where the data generated arose from pre-determined tasks which were fixed for all subjects), and in variable (unconstrained) tasks, where the precise data generated were directly under the control of the subject. The tasks of specific relevance here were as follows:

- *Task 1:* Subjects were asked to copy a list of defined words (43 words, 243 characters). The target text chosen was the following: 'The communication method: subroutine call or method invocation will not exit until the next invoked computation has been terminated. Asynchronous message passing, by contrast, can result in a response arriving a significant time after the request message has been sent through the net.' This text was chosen to encompass execution of an approximately even distribution of the most common digraphs and trigraphs occurring in the English language, providing a rich data generation environment.
- *Task 2:* Subjects were asked to look at a visual scene and to write a description of this in their own words. This visual stimulus was constructed from components intended to convey a positive and generally 'happy' message.
- *Task 3:* This was very similar to Task 2, but in this case the visual stimulus was intended to convey more of a 'sad' message.
- *Task 4:* Subjects were asked to copy a specified list of words (10 words and 50 characters) but had to do so within a time span of a maximum of 10 s. A count-down timer was prominently displayed during task execution. The chosen text formed the sentence 'All questions asked by five watch experts amazed the judge', which encompasses all the letters in the English alphabet.

All the writing samples were captured using a digitising tablet with a paper overlay to provide familiar feedback during the writing process.

In this paper we initially investigate just two of the possible higher-level states of interest, in this case the question of whether or not a subject is ‘happy’ or otherwise, and whether or not a subject is experiencing ‘stress’ or otherwise. To provide some ground truth index, we adopt the simplest possible approach to ascertaining this information, which is to ask each subject to indicate his/her degree of happiness and stress at the time of carrying out the task by assigning a ‘score’ between 1 (very unhappy etc.) and 10 (very happy etc). This is the sort of approach often used in basic experimentation [18], and is closely related to the use of the Likert Scale frequently adopted in psychometric testing (which, although accepted to be a somewhat imprecise method, nevertheless provides a degree of comparability with other studies in this broad area), although there are clearly other ways in which this sort of ground truth can be obtained. For example, in other experimental environments (e.g. in collecting samples of different facial expressions), it is quite common to use professional actors to simulate the required states of interest [9, 19], but these are usually states reflected entirely through a visible expression, which may not relate directly to a genuine inner emotional response, and this makes such an approach unsuitable here. We should also note that work is often reported where the emotion category is assumed to be directly implied by the task executed by subjects [20], while we (importantly) make no such assumption and record emotional state independently of task specification. Although not formally normalised, we address the problem of subjectivity and differing perceptions of degree of emotion in two ways. First, since we ask subjects to record their state at each test event, there is a degree of self-normalisation within the data. Second, by banding responses, and by introducing thresholding which includes a ‘no-assignment’ zone, we buffer the results from over-sensitivity to the thresholding process. We discuss this in greater practical detail elsewhere in the paper.

We accept that there are a wide range of opinions on how to define the ‘emotions’ we are adopting as the focus of our study, and that different disciplines will generate different definitions. Indeed, the whole question of what constitutes an ‘emotion’ is itself a matter of some debate. Although in future studies we believe it will be important to integrate input from other disciplines, in this initial work we have taken an approach, and used broad methodologies, found in other comparable studies reported, reflecting the fact that our study is intended to provide an explicit biometrics/forensics perspective.

By using this data collection protocol, experimental data were collected from 100 subjects (one sample each per task), with a 55:45 gender balance (male:female) and a spread of ages across three specified age groups of ‘<25’, ’25–60’ and ‘>60’, biased towards the lower end of the range, reflecting the conditions under which our subjects were recruited. The subjects were drawn from the student population and members of staff at the University of Kent in the UK and also from the general public and, although not screened in any way, represented a good mix of cultural and linguistic backgrounds. Naturally, we would have preferred to have a larger database, but 100 subjects was the limit of what was possible with the resources available. Nevertheless, if we compare other reported studies with similar objectives (although these are all working with a different biometric modality, namely keystroke analysis), we note that database sizes can be as small as 12 subjects, with the largest containing only 50 subjects (see, e.g. [11, 13]). Our database, despite its limitations, therefore represents the only data source available in this specific area and is twice the size of anything used in studies closest to those having any comparability with ours. Our database also provides handwriting samples collected under very different writing conditions, allowing us to evaluate the effects of these fundamentally different writing tasks, while also addressing some of the perceived weaknesses found in the other datasets referred to.

4 Experiments and results

4.1 Benchmarking performance

We begin by determining some basic predictive performance characteristics using the data acquired (based on the initial work reported in [4], of which this paper is an invited extension) and will then go on to develop the ideas further. We treat each experimental task as a separate event, and do not generally mix data across tasks.

Rather than try directly to model human-based assessment of relevant states, as we have in previous studies in a different problem domain [15], here we simply aim to classify subjects as ‘happy’ or ‘not happy’ and ‘stressed’ or ‘not stressed’, and we thus regard the prediction process as a two-class problem. However, our ground truth is related to a graduated scale (1–10) which, we acknowledge, is to an extent subjective and is non-normalised. Hence, for this study, we convert this scale to a two-category state population, by means of simple thresholding. However, to compensate for the subjectivity inherent in the process, we take scale points 9–10 to represent a ground truth state ‘happy’ and ‘stressed’ in the respective indicative scales, and scale points 1–7 to represent ‘not happy’ and ‘not stressed’. We discard subjects recording scores of 8 to increase the likelihood that the scores of the subjects included genuinely reflect a distinction between the two broad states of interest (and we also investigate in Section 4.2.1 the effect of varying thresholding boundaries).

After defining labels in this way, each handwriting sample is processed to form a representative template. First, 12 simple features (which the literature (see, e.g. [21–23]) shows are commonly used in handwritten data processing experiments) and which are shown in Table 1, are extracted from each handwriting sample, and are then mean and variance normalised.

Following this pre-processing of the handwritten data, classification is performed by using the simple K-nearest neighbour (K-NN) (with $K = 1$), incremental reduced error Prunung (Jrip) and support vector machine (SVM) classifiers using a leave-one-out cross validation methodology. For this process the Weka [24] software is used with default settings. A description of the classifiers may be briefly summarised as follows:

- *KNN classifier:* The KNN [25] algorithm defines a simple classifier which does not require a sophisticated training phase. Its only requirement is that the labels of classes represented by the samples are available. Thus, to find the nearest neighbour, an appropriate distance metric (simple Euclidean distance is used in this study) is calculated between each test sample and all training samples. The class label corresponding to the minimum distance in this

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Table 1 Extracted features

Feature number	Feature type	Feature description
F1	dynamic	the total time taken to execute the handwriting
F2	dynamic	average pen velocity in *x* direction
F3	dynamic	average pen velocity in *y* direction
F4	dynamic	average pen acceleration in *x* direction
F5	dynamic	average pen acceleration in *y* direction
F6	static	standard deviation of *x* coordinate values
F7	static	standard deviation of *y* coordinate values
F8	static	vertical centralness of the handwriting
F9	static	horizontal centralness of the handwriting
F10	dynamic	Pen Azimuth
F11	dynamic	Pen Altitude
F12	dynamic	Pen Pressure on capture tablet

list determines the assigned class label (since $K = 1$ in this study).

- *SVM:* The SVM [26] classifier approach finds a decision surface or a hyperplane between two point classes determined by certain points of the training set (support vectors). The decision surface/hyperplane shows the maximum distance between two classes. According to this hypersurface the training samples are divided into positive and negative groups, and then the classifier assigns the test sample to one of the two distinct classes.
- *Optimised incremental reduced error prunung:* This algorithm [27] uses a set of rules which, one by one, are tested to check whether a rule matches, all samples related to that rule then being deleted. This process is repeated until there are no more samples or the algorithm returns an unacceptable error. This algorithm uses a delayed pruning approach to avoid unnecessary pruning, resulting in a JRip procedure.

Prediction accuracy is evaluated for each task individually for both the 'happy' and the 'stressed' emotions. It is important to note that this prediction is person-independent since only one sample from each individual is available in each task. The results are presented in Figs. 1*a* and *b* for the 'happy' and for the 'stressed' prediction outcomes, respectively.

Considering the case of the 'happy' prediction, the best prediction performance is achieved by using the SVM classifier, with accuracies approaching 80% across all tasks. Jrip also achieves similar performance rates but is more task dependent, whereas the lowest and most variable accuracies observed between tasks are obtained with the *K*-NN classifier.

Considering the case of the 'stressed' prediction, again more consistent performance is achieved with the SVM classifier, which returns close to 70% accuracy. Moreover, a similar trend is observed with the Jrip and *K*-NN classifiers, where accuracy is again more variable between tasks.

Even these initial results raise some very interesting issues and possibilities and suggest that the following observations are applicable to the goal set out at the start of the paper:

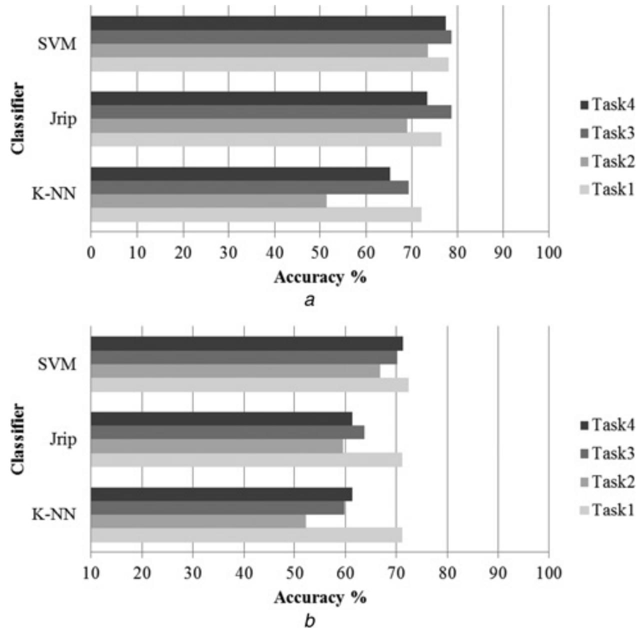

Fig. 1 *Accuracy of*

a 'Happy' prediction
b 'Stressed' prediction

- There is practical evidence that it is possible to identify 'higher level' state, reflecting the emotional disposition of an individual, based on an analysis of a fragment of that individual's handwriting.
- We can see that, unsurprisingly, the predictive capability observed will depend to an extent on the choice of classification infrastructure.
- The nature of the writing task will influence the prediction accuracy likely to be achievable. For example, a fixed task (Task 1), where the target writing has been chosen to embody a rich density of different elements which particularly characterise the language structure itself offers the likelihood of better performance than for unstructured tasks (Task 2 and Task 3).
- Within the set of unstructured tasks (Task 2 and Task 3), where the written content is uncontrolled and therefore relatively unpredictable, there is a greater variability in predictive accuracy. More specifically, one obvious uncontrolled factor here concerns the amount of handwritten text available. We have considered this further. Fig. 2 shows the distribution of fragment lengths (measured in terms of the number of words written) in the individual responses for Task 2 for the 'happy' and the 'stressed' emotions, broken down into the group of responses for which a correct prediction of assigned group was made, and for those where an incorrect assignment was made. We can see that the results suggest that there is some notional 'optimum' of the fragment length to work with. For smaller fragment sizes the increasing richness of information afforded by increasing sample size results in improving prediction accuracy.
- The differences between different tasks are somewhat reduced if a sufficiently powerful classifier is adopted for the prediction processing.

It is worth also noting a particular point about our methodology. The tests were framed in a way which, it might be expected, could emphasise one emotion of interest

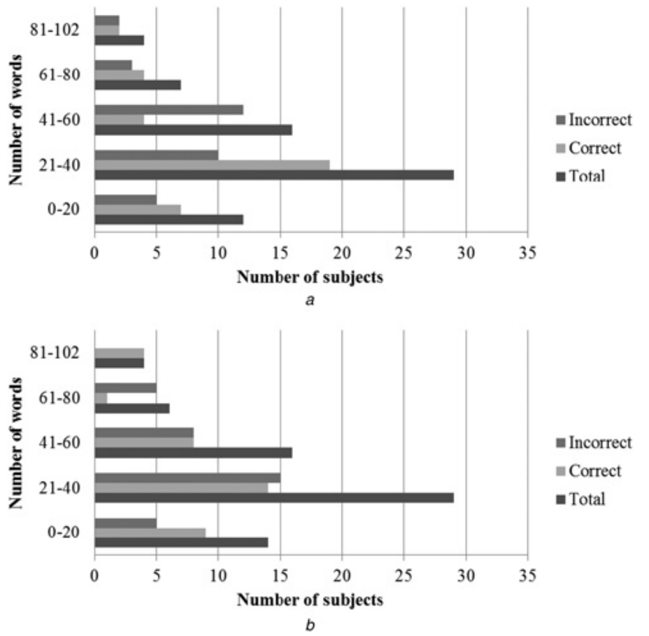

Fig. 2 *Number of words against correct prediction of assigned group for Task 2*

a 'Happy' emotion
b 'Stressed' emotion

over another. For example, Task 2 might nudge a participant towards a more 'happy' frame of mind, whereas Task 4 might similarly modestly increase the 'stress' score. We made no assumptions about this, however, and our approach was designed primarily to encourage both a spread of responses without overt intervention and a spread of self-evaluation response indices. Fig. 3, recording the score distributions for all tasks and for the 'happy' and 'stress' emotions, does show that this is the case, although not to any marked degree, and this might suggest that some further consideration of the optimality of this methodology would be beneficial.

Hence, even a first analysis of our results shows a rather encouraging outcome while, moreover, pointing constructively towards some important areas for further work. Especially in relation to the applicability of this type of analysis for scenarios in forensics, however, it is important to look in a little more detail at what these data suggest.

4.2 Further analysis

Moving beyond this initial analysis, then, it is possible to develop these ideas further, and so we briefly investigate here some issues relating to other factors of relevance in the approach adopted, considering specifically

- The assignment of category labels to the characteristics of interest in our predictive study in relation to the designation provided by subjects.
- The nature of the feature set used for predictive purposes.
- The potential for inter-task training/test dataset crossover in carrying out the prediction operation.

Although the first of these will provide some insight into an area where our methodology is not completely objectively defined, the second two issues will provide a greater understanding of the potential effectiveness of our predictive techniques in some operational environments likely to be encountered in forensic analysis applications.

4.2.1 Category assignment: In our initial results we assigned the 'happy' label to subjects who recorded a score of 9 or 10 on the rating scale, and 'unhappy' to those recording a score from 1 to 7, while excluding subjects who rated themselves at the 8 level, thus providing a clear distinction between the two groups. However, Table 2 shows some results for the SVM classifier when different category boundaries are adopted. Here we analyse the data for Task 1, where the data exhibit the maximum imposed textual richness. The two groups Gr1 and Gr2 here correspond, respectively, to the subject groupings (test sub-populations) for those rated below a lower cut- off point (≤ 7 in the previous results) and above an upper cut-off point (≥ 9 in the previous results). Row 1 shows the prediction accuracy for lower and upper boundaries of 5 and 7, respectively, and Row 2 for boundaries of 6 and 8. Row 3 shows, for comparison, the performance figures for the boundaries originally used.

We see that extending the boundaries for the 'happy' category (i.e. including in that category those who assigned themselves a rating score of 7 or above) and decreasing the cut-off for the 'unhappy' category' to a score of 5 or less appears to increase the predictive capacity substantially, and this suggests that it is important to interpret the rating scale appropriately in this type of analysis. This, however, is especially difficult with limited data, where an objective

Table 2 Accuracy of 'happy' prediction with respect to different category boundaries

Boundaries	Gr1	Gr2	Accuracy %
lower = 5, upper = 7	9	75	89.3
lower = 6, upper = 8	25	47	65.3
lower = 7, upper = 9	53	15	77.9

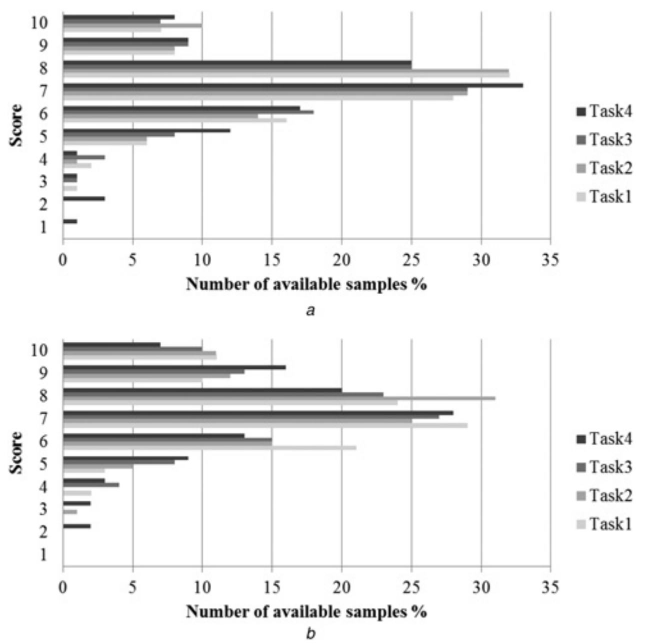

Fig. 3 *Score distributions for all tasks and for*

a 'Happy' emotion
b 'Stressed' emotion

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ground truth is very difficult to establish. Changing the upper and lower boundaries to those corresponding to Row 2 in Table 2, on the other hand, is seen to substantially reduce the predictive performance achievable.

However, there is another important factor to consider here. In Table 2 the entries under Gr1 and Gr2 show the number of subjects categorised within the two groups on the basis of the category boundaries chosen. It can be seen that the boundary choice has a very significant effect on the balance of numbers of subjects in the two groups, and this too could have a marked impact on the predictive performance measured. On this basis, we note that the results shown in Row 2 are those obtained when the numbers are closest to being in balance (although even here there is still some discrepancy), perhaps suggesting that these conditions better reflect likely performance in practice. It is clear, though, that this is a complex issue, and the limitations of data availability will need to be addressed before more meaningful conclusions can be drawn.

4.2.2 Nature of the feature set: One of the important issues when analysing handwritten data is that the way in which a handwriting sample is collected will determine the nature of the features which can be extracted. In biometrics, these are broadly characterised as either static features (related to overall shape and appearance information, available from off-line or on-line sample acquisition) or dynamic features (time-related information relating to execution of the handwriting and only available from an on-line acquisition process). It is also possible to define a number of 'pseudo-dynamic' features (dynamic information inferred from the static data), though this type of information is more commonly used in human forensic analysis than in specifically biometric processing.

From our present perspective, although all our handwriting samples were collected using dynamic capture (via a digitising tablet), it should be recognised that, at least currently, most typical forensic analysis scenarios are likely to involve off-line capture of handwriting fragments. Thus it is important to investigate the implications of this for the predictive techniques being considered here (although our technique has many other potential applications where the full feature set can be made available).

For illustrative purposes we again consider the task of predicting the 'happy' state, but here, using the K-NN classifier as an illustrative example, we compare prediction performance for different feature groupings. Specifically, we consider the performance obtained when we use only the static features found in the feature pool described earlier (there are just four of these), when we use only the dynamic features (there are eight of these), and compare these scenarios with the case already reported using all twelve features, both the static and the dynamic.

Table 3 shows the comparative performance achieved in these circumstances for Tasks 1, 2 and 3, namely the fixed task, where all subjects write the same text, and two tasks where (different) free-format and unconstrained text is produced by each subject.

If we consider Task 1, we see that the prediction performance follows a pattern which might be expected. The use of only static features reduces the predictive power of the system, while using the dynamic features alone returns a performance level only slightly below that obtained when all the features are used. As would be expected [28, 29], the increased richness of the dynamic properties of handwriting execution has a significant impact on how well the features embody the characteristics of the user compared with the static features. However, two observations can be made about the implications for forensics-oriented applications. First, it should be noted that the set of static features used here comprises only four features which are, moreover, relatively simple features. Second, given such a small and rudimentary set of features, the fact that we are able to achieve a predictive accuracy of more than 57% is rather encouraging. Indeed, we can take this further, simply by extending the static feature set. For example, we added three further such simple static features (two relating to x- and y-axis local pen excursions, respectively, and one relating to x/y perspective), resulting in an increase in predictive accuracy from 57.3 to 64.7%. There is clearly scope for more rigorous development of the feature set, perhaps coupled with some optimisation based on assessing feature correlations.

Table 3 Accuracy of 'happy' prediction with respect to different feature types

Task type	Accuracy %		
	All features	Static features	Dynamic features
task 1	72.1	57.4	69.1
task 2	51.5	54.4	55.9
task 3	69.3	73.3	65.3

In the free-format tasks (Task 2 and Task 3), the picture is much less clear. Although we see there is a possibility of achieving even better static-based performance, there is a lack of consistency in the results compared with what we observed in Task 1. This is simply because in these unconstrained tasks, the data available from each subject vary widely, and so performance is related more directly to subject-specific response. Thus, while specific conclusions are difficult to draw here, there are very positive pointers about how to develop further a strategy for forensic analysis even in such circumstances.

4.2.3 Inter-task operation: We have shown earlier that attainable predictive performance can vary depending on the nature of the task domain in which the data are collected and, specifically, we have considered situations where all subjects write the same text and, alternatively, where the task is free-format, the text generated being entirely determined by the writer and the task content specification. In a forensics context, it is likely that a test sample and a reference sample might often have been generated under rather different conditions (a forensic investigation cannot always control the circumstances under which a writing sample becomes available). Here we investigate the effect of analysing a test sample for predictive purposes when the prediction system has been trained under different conditions.

Again we look at a prediction of the 'happy' characteristic and use the K-NN classifier as our processing platform. Table 4 shows the prediction performance when training takes place using data from a particular experimental Task (the column labelled 'Train') but the performance is assessed using data from a different Task (as noted in the column labelled 'Test').

These results again point to the benefits of working with the rich data obtained in Task 1, the task where all subjects provide the same task content. When such data are used in testing, the predictive performance observed is generally

Table 4 Accuracy of 'happy' prediction with respect to different testing and training sets

Test	Train	Accuracy %
task 1	task 1	72.1
	task 2	69.1
	task 3	63.2
task 2	task 1	55.9
	task 2	51.5
	task 3	61.8
task 3	task 1	66.7
	task 2	50.7
	task3	69.3

substantially better than in other cases, and depends to a lesser degree on what samples were used in training. Again, we see that performance is considerably more variable when the system is tested on the free-format data (consistent with what we have observed elsewhere) irrespective of what training data were used. There may be important lessons here for the potential applicability in forensic analysis, where we may have limited choice over the nature of the samples to be used in testing. Nevertheless, we can begin to see how these results can support the development of a possible strategy for establishing and perhaps optimising confidence in prediction from handwritten fragments.

5 Conclusions

We have demonstrated that prediction of higher level mental state based on handwritten fragments is a viable possibility, but we have also shown that many interrelated factors, from the nature of the data and the task environment through to implementation issues, will have a significant effect on the confidence which can be ascribed to the predictive outcomes obtained.

The paper has also established some basic steps towards developing this type of work further, and has presented results which indicate some potentially fruitful avenues to pursue.

The link between the predictive capacity of the analysis of handwritten fragments and its possible value in the context of forensic analysis, using the predictive approach described above, is very clear. Our initial discussion recognises some limitations on what we can conclude at this stage, and it is clear that there are a number of interrelated factors which need to be disambiguated as the work proceeds. On an overall view, however, the basic picture to emerge from this work is a very positive one.

More specifically, what we learn from this study is that the increasing understanding we are developing will help us to work towards implementing a procedure which can enhance prediction accuracy. This will take into account factors such as the nature and content of writing samples available, the relation between 'reference' data and test sample presented for analysis, and the features which can be extracted and utilised in a particular application domain and test situation.

This leads us to point to a number of issues which will provide the principal focus in developing this study in the future. For example, some of our priorities will include

- Developing a better understanding of the relation between individual features and predictive capability.
- Refining the way in which we can establish reliable ground truth markers for the type of data of interest here. Despite the fact that we have taken an approach similar to that commonly adopted, this is an area where exploring alternatives may prove particularly productive.
- Identifying approaches to optimising performance in relation to particular task environments.

All of these things, however, depend most of all on acquiring more data (both in quantity and in diversity) on the basis of which to develop a more comprehensive and more rigorous analysis. The unique database we have acquired to date has been a vital prerequisite to establishing the framework of the study we have reported here, and we need now to develop this further to provide the data required to probe more deeply into this important issue of predictive capability.

Our study has taken some clear steps towards establishing an approach to prediction of important human characteristics which can provide valuable and desirable information in a wide range of application areas, and we aim in our future work to build on this foundation to provide further new insights in an area where reported existing work has not so far been extensive.

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